

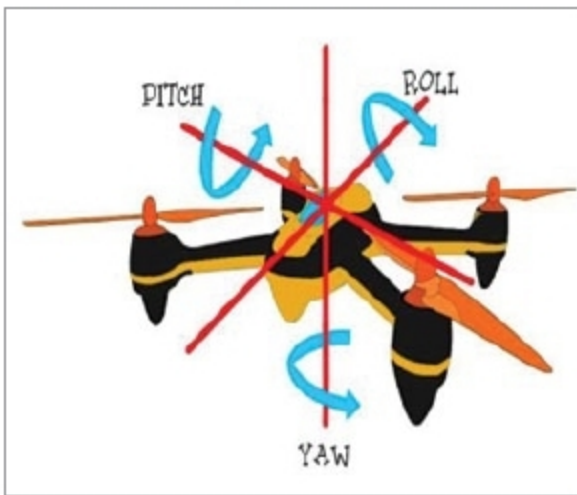


Fig. 5: Motion controls of the quadcopter (Credit: www.droneeybee.com)

pushing the left stick to the left (to turn left) or to the right (to turn right).

Pitch and roll. Pitch is the movement of the quadcopter, either forward or backward. Roll is the movement of the quadcopter side-to-side, either left or right.

A quadrotor adjusts its pitch or roll by applying more thrust to one rotor and less thrust to its diametrically opposite rotor.

On most quadcopters, roll is controlled by pushing the right stick on the transmitter to the left (to roll to the left) or to the right (to roll to the right). And on most quadcopters, pitch is controlled by pushing the right stick on the transmitter up (to move forwards) or down (to move backwards).

The drone ambulance helps in assisting ambulances, which help in saving human lives. The drone travels through air and thus faces no traffic. In this project, three different paths were considered. Observation proved that the drone was able to reach the spot early.

Real-time health parameters of the patient that are transmitted by ZigBee and observed in LabVIEW software, those results and graphs are presented in Figs 6, 7 and 8. Results include measured data of such sensors as ECG, temperature and heartbeat.

- **ECG sensor data:** ECG electrodes were attached to the left and right arms and the right leg of the patient. Electrodes convert heartbeat to electrical signal and, thus, ECG sensors can measure continuous heartbeat and give heart rate data. Measured data was sent through ZigBee to the ambulance and data was observed in LabView. This data

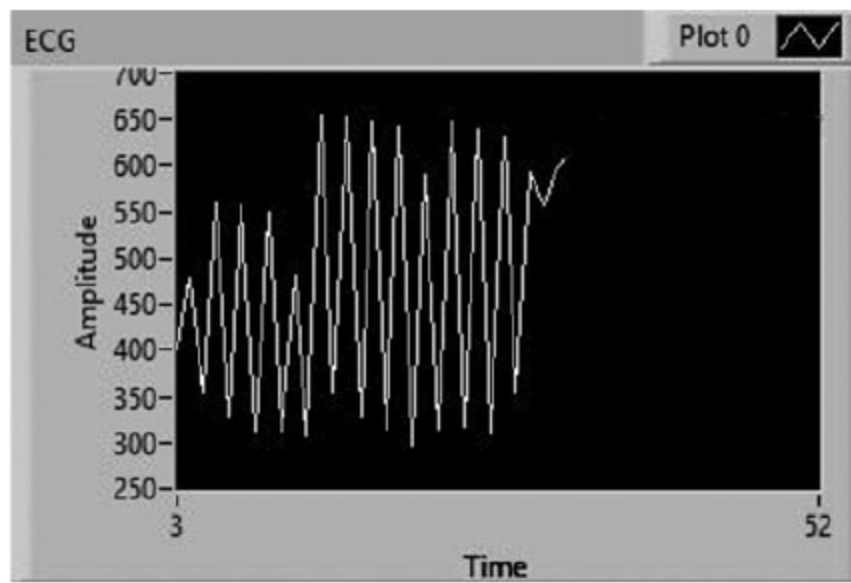


Fig. 6: Observed ECG sensor data



Fig. 7: Observed heartbeat sensor data

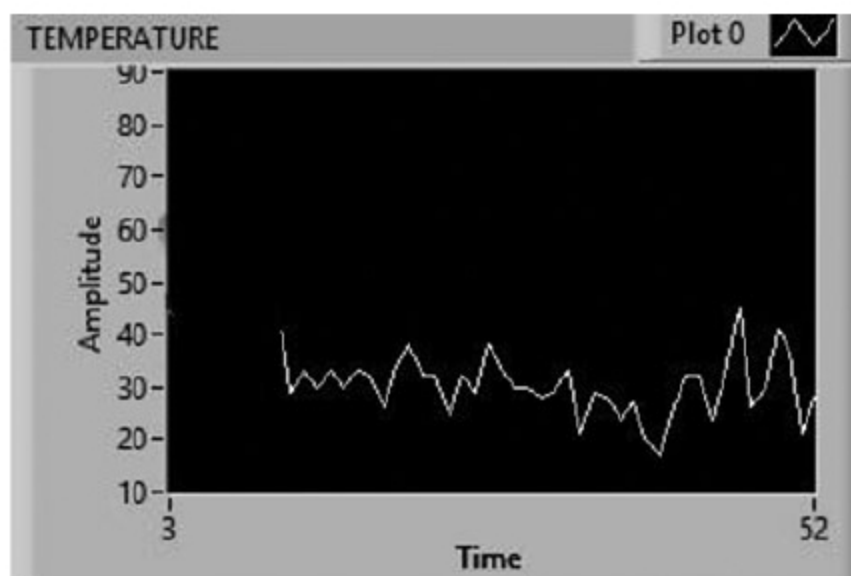


Fig. 8: Observed temperature sensor data

shows the condition of the heart, that is, rate of the heart. Analogue value of the heart rate can be seen on the LCD display screen.

- **Heartbeat data:** Pulse rate or heartbeat sensor measures the heartbeat of the patient. Data was sent through ZigBee to the ambulance. Observed data in LabView is shown in Fig. 7. The graph shows heartbeat measurement as 72. Heartbeat in the shown graph varies from 69 to 72. Data can also be seen on the LCD

display screen in the medbox.

- **Temperature data:** The thermistor measures temperature of the sensor, and data was transmitted through ZigBee to the ambulance. Data thus observed in LabView is shown in Fig. 8. Temperature shown measures approximately 30°C. Atmospheric conditions and other factors might cause some variation in data, but data shown on the LCD can also be used to estimate this value.

These results represent data of the patient, and this data can be sent to the ambulance where it can be analysed by the doctor so that he or she can evaluate the situation better and take other pre-medication choices.

Data taken essentially proves beneficial for the patient and, hence, the purpose of the drone ambulance, which is to assist ambulances in saving human lives.

The future

Significantly further headways should be

possible. Automation can be made self-ruling. GPS module can be utilised to obtain area of the objective, and machine learning can be utilised to assist with automation by calculating the briefest way to achieve the goal.

A camera can be actualised in automation with the goal that present circumstances can be seen by the specialist in the emergency vehicle. A significantly heartier automation can be created for completing this particular task. **EFY**